CROCO Benchmark Report

REGIONAL › BENGUELA_LR · grid 60 × 40 × 32 · 30 days per run · 2026-10-24 17:35

Configuration	cppdefs.h block REGIONAL, case BENGUELA_LR
Grid (param.h)	LLm × MMm × N = **60 × 40 × 32** (active branch REGIONAL › BENGUELA_LR, line 10)
Machine	vm: Intel(R) Xeon(R) Processor @ 2.10GHz; 2 physical / 2 logical CPUs
Cores used	16 (CPU 0-15), up to 16 simultaneous runs
Run directory	/home/claude/mock/case
Simulated period	30 days per run; original croco.in dt = 3600 s, NDTFAST = 60 (fixed)

1. Key results

RECOMMENDED DT

1200 s

3600 / 3 steps per hour; largest stable 1790 s

RECOMMENDED CORES · MPI

8 cores

layout 4x2, E = 0.506

RECOMMENDED CORES · OPENMP

4 cores

layout 1x4, E = 0.581

FASTEST RUN

MPI 12 cores

6 × 2, 1.5 s per 30 days

BAROTROPIC CFL AT RECOMMENDED DT

0.567

advective 0.414; first BLOW UP at 1800 s

SERIAL REFERENCE T(1)

7.4 s

1 core, no OpenMP/MPI, dt = 1200 s

- **Recommended time step: 1200 s** = 3600 / 3, the largest integer dt that divides 3600 without exceeding the largest stable dt. All times and efficiencies in this report were measured with it.
- **Largest stable dt found:** 1790 s (NDTFAST = 60, barotropic dt = 29.83 s).
- **Stability limit** between 1790 s (stable) and 1800 s (first BLOW UP).
- **MPI:** up to **8 cores** (layout 4x2, efficiency 0.506 > 0.5).
- **OPENMP:** up to **4 cores** (layout 1x4, efficiency 0.581 > 0.5).
- **Fastest configuration:** MPI with 12 cores, layout 6 × 2 (NP_XI × NP_ETA), 1.5 s per 30 days (71760.8 simulated days per wall-clock hour), measured alone on the machine.

2. Time-step search (serial run)

All runs of this stage use the serial build (1 core, OPENMP and MPI undefined). Starting from the reference dt (600 s), dt is doubled while the run is stable; after the first BLOW UP the next dt is the midpoint between the last stable dt and the last BLOW UP, and the search stops after two BLOW UPs in a row. Rules: a dt larger than one that blew up is never tested, and when going down the search never goes below a stable dt whose half is also stable.

Table 1. Decision sequence of the time-step search.

Step	Phase	dt [s]	Result
1	up	600	stable
2	up	1200	stable
3	up	2400	BLOW UP
4	bisect	1800	BLOW UP
5	bisect	1500	stable
6	bisect	1650	stable
7	bisect	1725	stable
8	bisect	1762	stable
9	bisect	1781	stable
10	bisect	1790	stable

Table 2. Stability and Courant numbers for every dt tested (★ = recommended dt). Internal-wave CFL estimated with an assumed c_1 = 2.0 m/s.

dt [s]	NDTFAST	dt_fast [s]	Status	CFL barotropic	CFL internal	CFL advective	Best time
600	60	10.00	OK	0.284	0.768	0.207	14.6 s
1200 ★	60	20.00	OK	0.567	1.537	0.414	1.5 s
1500	60	25.00	OK	0.709	1.921	0.517	6.1 s
1650	60	27.50	OK	0.780	2.113	0.569	5.6 s
1725	60	28.75	OK	0.815	2.209	0.595	5.7 s
1762	60	29.37	OK	0.833	2.256	0.608	5.6 s
1781	60	29.68	OK	0.842	2.281	0.614	5.7 s
1790	60	29.83	OK	0.846	2.292	0.618	5.8 s
1800	60	30.00	BLOWUP	0.851	2.305	0.621	–
2400	60	40.00	BLOWUP	1.134	3.074	0.828	–

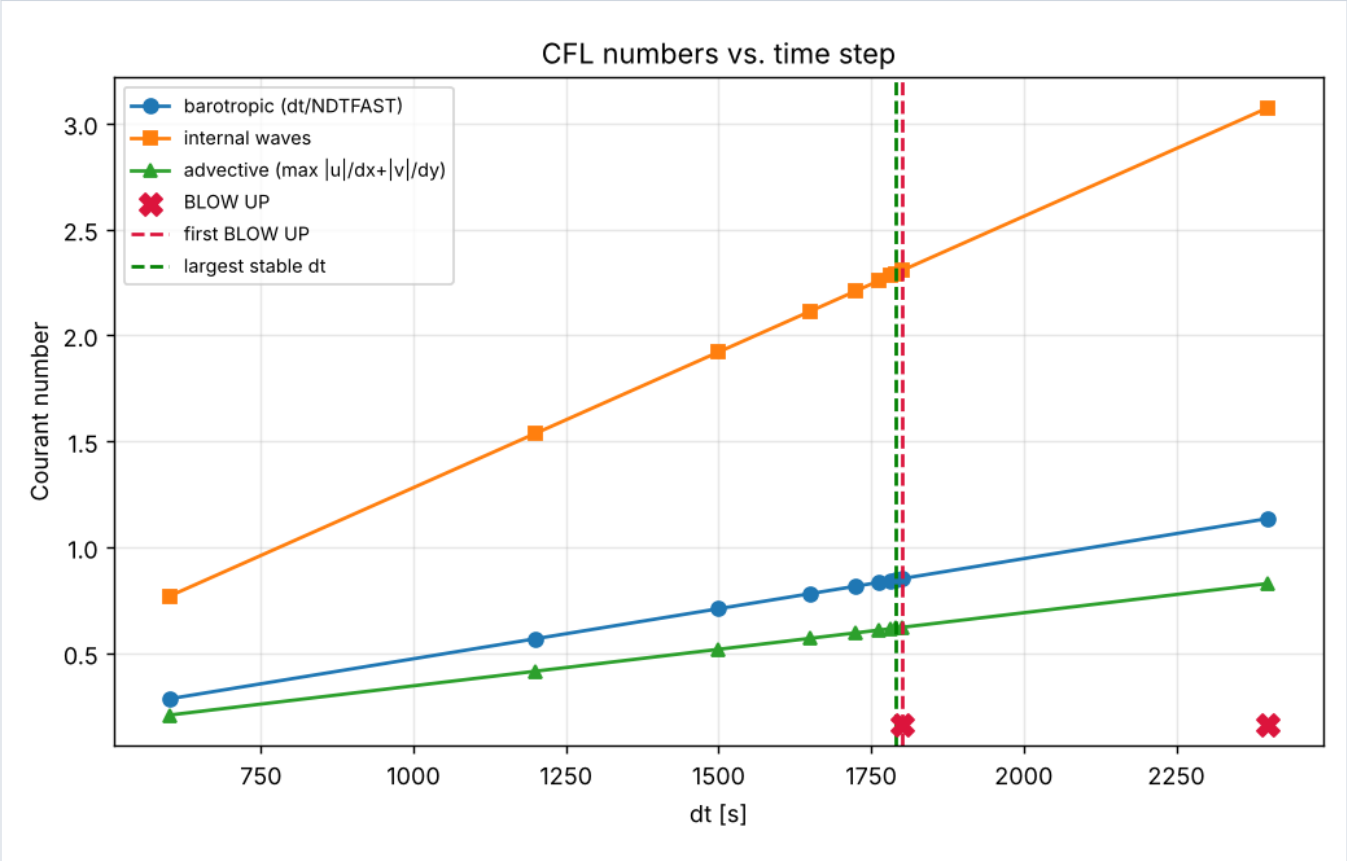


Figure 1. *Courant numbers versus time step; the dashed lines mark the largest stable dt and the first BLOW UP.*

3. Core-count exploration

OpenMP and MPI start at 12 cores. For each core count every valid nx × ny decomposition is run and the fastest is kept. With E(n) = T(1) / (n · T(n)): if E > 0.5 the next test uses 2 more cores, otherwise 2 fewer; each mode stops when the direction reverses or the core limit is reached.

Table 3. OPENMP exploration path (stop: direction reversed at 4 cores).

Cores	Best layout	Time	Efficiency E	Decision
12	4x3	1.9 s	0.329	E ≤ 0.5 → −2
10	5x2	2.0 s	0.362	E ≤ 0.5 → −2
8	4x2	2.2 s	0.421	E ≤ 0.5 → −2
6	2x3	2.5 s	0.493	E ≤ 0.5 → −2
4	1x4	3.2 s	0.581	E > 0.5 → +2

Table 4. MPI exploration path (stop: direction reversed at 8 cores).

Cores	Best layout	Time	Efficiency E	Decision
12	6x2	1.5 s	0.409	E ≤ 0.5 → −2
10	5x2	1.7 s	0.444	E ≤ 0.5 → −2
8	4x2	1.8 s	0.506	E > 0.5 → +2

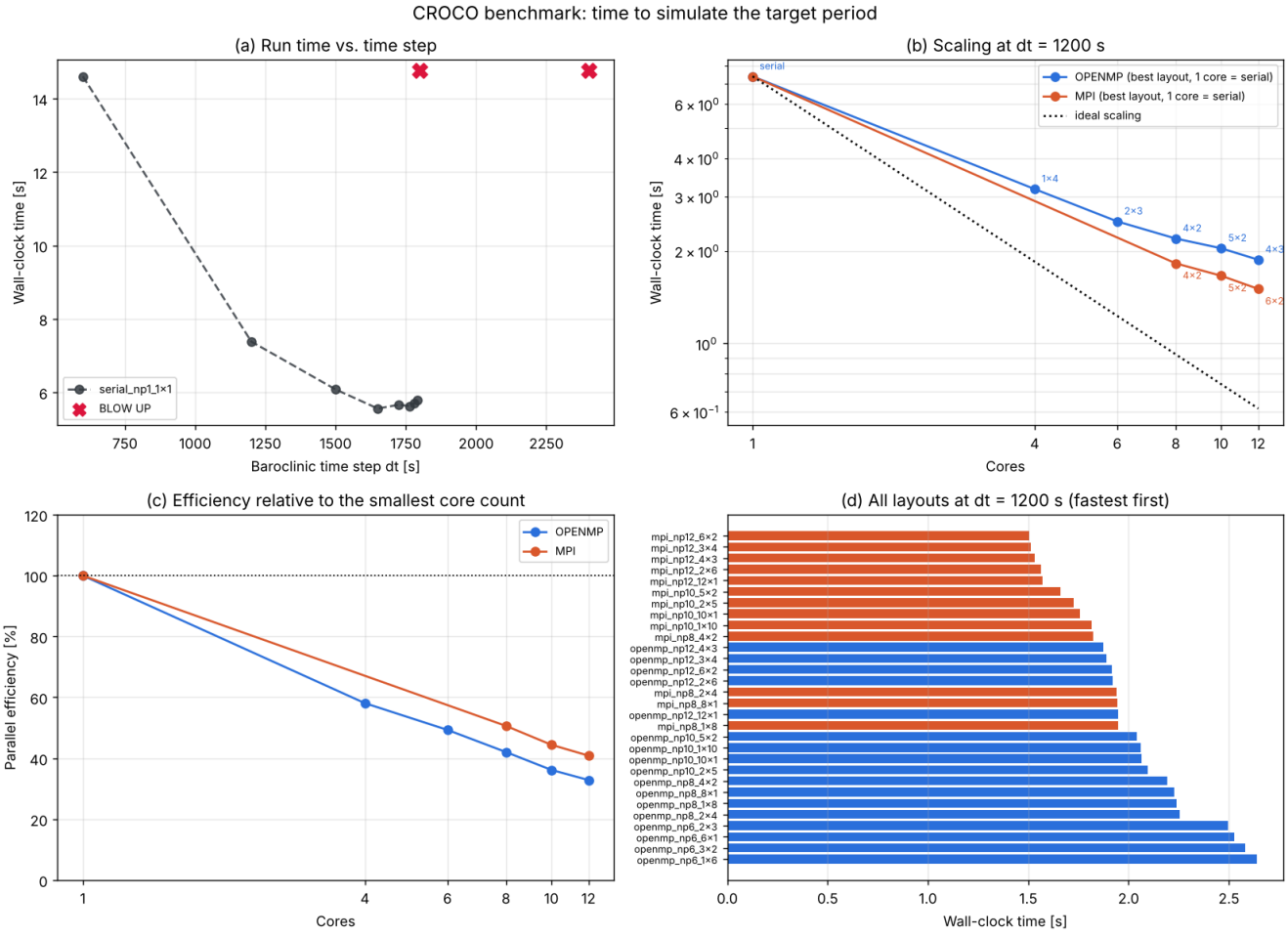


Figure 2. *(a) run time versus dt, (b) scaling and (c) parallel efficiency at dt = 1200 s, (d) every layout tested.*

4. Efficiency study (NLHPC format)

Format of the NLHPC efficiency study, at dt = 1200 s and 30 days simulated, with the best layout of each core count: **speed-up** S(n) = T(1) / T(n), **efficiency** E(n) = S(n) / n. T(1) is the serial run (1 core, built without OpenMP or MPI). Recommended: the largest core count with E > 0.5.

Table 5. MPI: recommended **8 cores**.

No. of cores	1	8	10	12
Layout	serial	4x2	5x2	6x2
Time [s]	7.38	1.82	1.66	1.50
Speed-up	1.00	4.05	4.44	4.90
Efficiency	**1.00**	**0.51**	0.44	0.41
Measured alone	yes	yes	yes	yes

Table 6. OPENMP: recommended **4 cores**.

No. of cores	1	4	6	8	10	12
Layout	serial	1x4	2x3	4x2	5x2	4x3
Time [s]	7.38	3.18	2.49	2.19	2.04	1.87
Speed-up	1.00	2.32	2.96	3.36	3.62	3.94
Efficiency	**1.00**	**0.58**	0.49	0.42	0.36	0.33
Measured alone	yes	yes	yes	yes	yes	yes

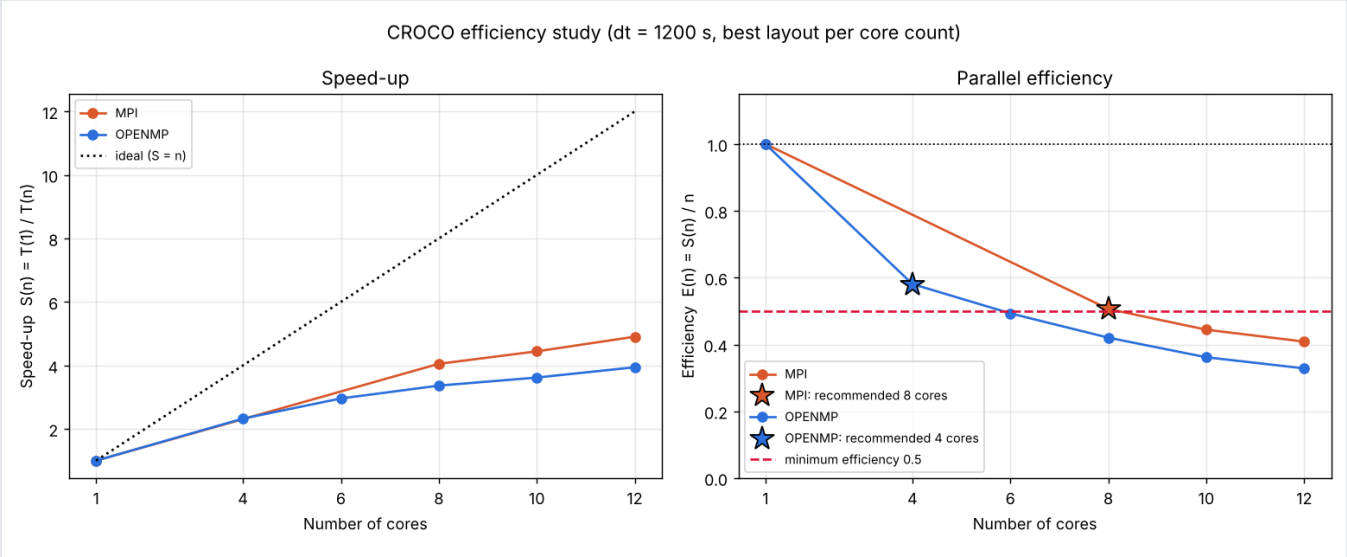


Figure 3. *Speed-up (left, with the ideal line) and parallel efficiency (right, with the threshold); stars mark the recommended core count of each mode.*

5. All layouts at dt = 1200 s

Table 7. Every layout at dt = 1200 s, fastest first; speed-up and efficiency relative to the fastest run with 1 core(s).

Mode	Cores	Layout	Time	Days/hour	Speed-up	Efficiency	Measured	Status
MPI	12	6 × 2	1.5 s	71760.8	4.90	41 %	alone	OK
MPI	12	3 × 4	1.5 s	71475.8	4.88	41 %	alone	OK
MPI	12	4 × 3	1.5 s	70404.2	4.81	40 %	alone	OK
MPI	12	2 × 6	1.6 s	69053.7	4.72	39 %	alone	OK
MPI	12	12 × 1	1.6 s	68746.0	4.70	39 %	alone	OK
MPI	10	5 × 2	1.7 s	65021.1	4.44	44 %	alone	OK
MPI	10	2 × 5	1.7 s	62500.0	4.27	43 %	alone	OK
MPI	10	10 × 1	1.8 s	61503.4	4.20	42 %	alone	OK
MPI	10	1 × 10	1.8 s	59536.9	4.07	41 %	alone	OK
MPI	8	4 × 2	1.8 s	59275.5	4.05	51 %	alone	OK

Mode	Cores	Layout	Time	Days/hour	Speed-up	Efficiency	Measured	Status
OPENMP	12	4 × 3	1.9 s	57692.3	3.94	33 %	alone	OK
OPENMP	12	3 × 4	1.9 s	57142.9	3.91	33 %	alone	OK
OPENMP	12	6 × 2	1.9 s	56367.4	3.85	32 %	alone	OK
OPENMP	12	2 × 6	1.9 s	56220.7	3.84	32 %	alone	OK
MPI	8	2 × 4	1.9 s	55698.8	3.81	48 %	alone	OK
MPI	8	8 × 1	1.9 s	55527.0	3.79	47 %	alone	OK
OPENMP	12	12 × 1	1.9 s	55498.5	3.79	32 %	alone	OK
MPI	8	1 × 8	1.9 s	55498.5	3.79	47 %	alone	OK
OPENMP	10	5 × 2	2.0 s	52941.2	3.62	36 %	alone	OK
OPENMP	10	1 × 10	2.1 s	52427.2	3.58	36 %	alone	OK
OPENMP	10	10 × 1	2.1 s	52350.9	3.58	36 %	alone	OK
OPENMP	10	2 × 5	2.1 s	51502.1	3.52	35 %	alone	OK
OPENMP	8	4 × 2	2.2 s	49225.2	3.36	42 %	alone	OK
OPENMP	8	8 × 1	2.2 s	48474.0	3.31	41 %	alone	OK
OPENMP	8	1 × 8	2.2 s	48214.3	3.30	41 %	alone	OK
OPENMP	8	2 × 4	2.3 s	47893.6	3.27	41 %	alone	OK
OPENMP	6	2 × 3	2.5 s	43303.9	2.96	49 %	alone	OK
OPENMP	6	6 × 1	2.5 s	42738.4	2.92	49 %	alone	OK
OPENMP	6	3 × 2	2.6 s	41844.2	2.86	48 %	alone	OK
OPENMP	6	1 × 6	2.6 s	40909.1	2.80	47 %	alone	OK
OPENMP	4	1 × 4	3.2 s	34005.0	2.32	58 %	alone	OK
OPENMP	4	2 × 2	3.2 s	33845.2	2.31	58 %	alone	OK
OPENMP	4	4 × 1	3.2 s	33374.5	2.28	57 %	alone	OK
SERIAL	1	1 × 1	7.4 s	14632.2	1.00	100 %	alone	OK

6. Run schedule

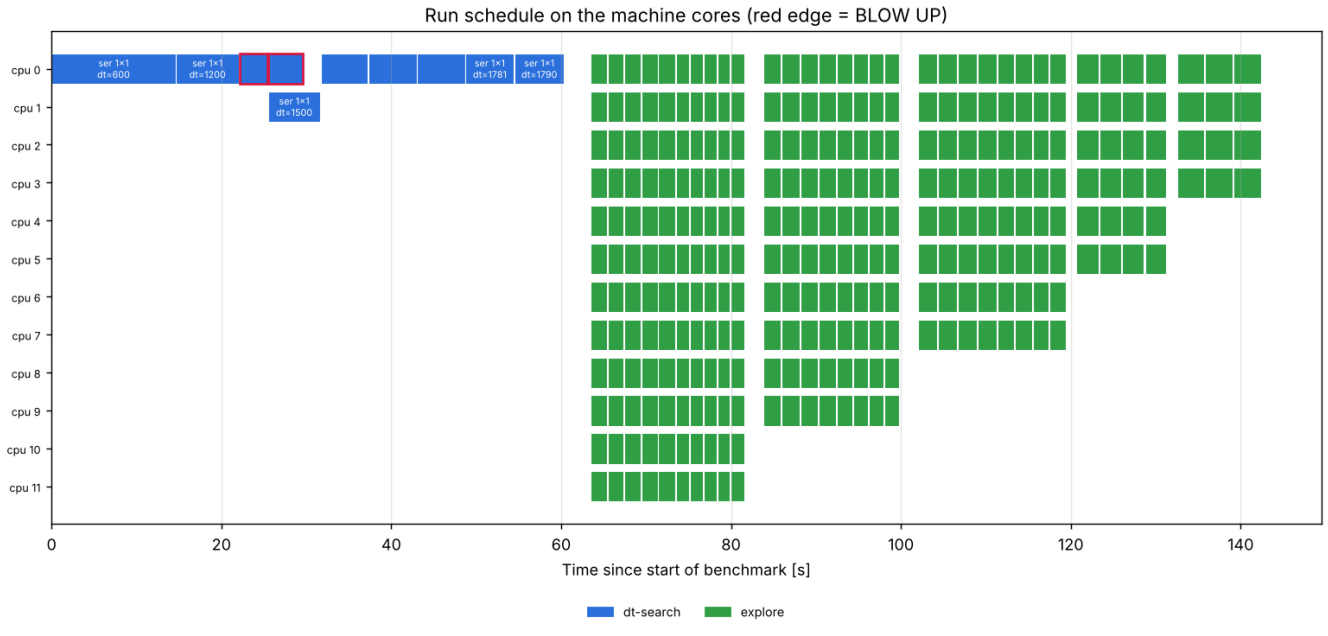


Figure 4. *Which run used which cores and when (red edge = BLOW UP).*

7. Methodology

- Every run starts from the original `cppdefs.h`, `param.h` and `croco.in`. `cppdefs.h`: `OPENMP` or `MPI` switched inside the active configuration block (both undefined for the serial build); `param.h`: `NP_XI/NP_ETA` (MPI) or `NPP/NSUB_X/NSUB_E` (OpenMP); `croco.in`: `dt` and `NTIMES = ceil(30·86400/dt)`. Output periods are rescaled to stay constant in seconds.
- Case and grid are not assumed: the preprocessor directives of `cppdefs.h` are evaluated and then used to find the `parameter (LLm0=…, MMm0=…, N=…)` line of `param.h` that is actually compiled.
- One executable per layout, compiled before the runs that use it, several at a time in sibling directories of `run_dir`; changing dt does not recompile. Every run works in its own directory (`work/<run>`) with its own `croco.in`, inputs linked read-only, and is pinned to its own block of cores (`taskset`, `OMP_PLACES`).
- Measured time: wall-clock time from launch to process end. BLOW UP is detected in the log (`BLOW UP` or `NaN`); the run is killed after a grace period and runs with a larger dt are cancelled.
- CFL barotropic = dt_fast·√(g·h)·√(1/dx²+1/dy²) (maximum over wet points, dt_fast = dt/NDTFAST); CFL internal = dt·c_1·√(1/dx²+1/dy²) with an assumed c_1; CFL advective = dt·max(|u|/dx + |v|/dy) from the history file.
- Stability over the simulated period does not guarantee stability under stronger forcing: for production, keep a margin (≈ 10–20 % below the largest stable dt).

8. Appendix: all runs

Table 8. Every run of the benchmark, in order.

Stage	Mode	Cores	Layout	dt	Rep	CPUs	Concurrent	Status	Time [s]	Sim. days	Notes
dt-search	serial	1	1×1	600	0	0	0	OK	14.6	30.000	
dt-search	serial	1	1×1	1200	0	0	0	OK	7.4	30.000	
dt-search	serial	1	1×1	2400	0	0	0	BLOWUP	3.3	18.000	
dt-search	serial	1	1×1	1800	0	0	1	BLOWUP	4.1	18.000	
dt-search	serial	1	1×1	1500	0	1	1	OK	6.1	30.000	
dt-search	serial	1	1×1	1650	0	0	1	OK	5.6	30.002	
dt-search	serial	1	1×1	1725	0	0	1	OK	5.7	30.008	
dt-search	serial	1	1×1	1762	0	0	1	OK	5.6	30.019	
dt-search	serial	1	1×1	1781	0	0	1	OK	5.7	30.013	
dt-search	serial	1	1×1	1790	0	0	0	OK	5.8	30.020	
explore	openmp	12	2×6	1200	0	0-11	0	OK	1.9	30.000	
explore	openmp	12	3×4	1200	0	0-11	0	OK	1.9	30.000	
explore	openmp	12	4×3	1200	0	0-11	0	OK	1.9	30.000	
explore	openmp	12	6×2	1200	0	0-11	0	OK	1.9	30.000	
explore	openmp	12	12×1	1200	0	0-11	0	OK	1.9	30.000	
explore	mpi	12	2×6	1200	0	0-11	0	OK	1.6	30.000	
explore	mpi	12	3×4	1200	0	0-11	0	OK	1.5	30.000	
explore	mpi	12	4×3	1200	0	0-11	0	OK	1.5	30.000	
explore	mpi	12	6×2	1200	0	0-11	0	OK	1.5	30.000	
explore	mpi	12	12×1	1200	0	0-11	0	OK	1.6	30.000	
explore	openmp	10	1×10	1200	0	0-9	0	OK	2.1	30.000	
explore	openmp	10	2×5	1200	0	0-9	0	OK	2.1	30.000	
explore	openmp	10	5×2	1200	0	0-9	0	OK	2.0	30.000	
explore	openmp	10	10×1	1200	0	0-9	0	OK	2.1	30.000	
explore	mpi	10	1×10	1200	0	0-9	0	OK	1.8	30.000	
explore	mpi	10	2×5	1200	0	0-9	0	OK	1.7	30.000	
explore	mpi	10	5×2	1200	0	0-9	0	OK	1.7	30.000	
explore	mpi	10	10×1	1200	0	0-9	0	OK	1.8	30.000	
explore	openmp	8	1×8	1200	0	0-7	0	OK	2.2	30.000	

Stage	Mode	Cores	Layout	dt	Rep	CPUs	Concurrent	Status	Time [s]	Sim. days	Notes
explore	openmp	8	2×4	1200	0	0-7	0	OK	2.3	30.000	
explore	openmp	8	4×2	1200	0	0-7	0	OK	2.2	30.000	
explore	openmp	8	8×1	1200	0	0-7	0	OK	2.2	30.000	
explore	mpi	8	1×8	1200	0	0-7	0	OK	1.9	30.000	
explore	mpi	8	2×4	1200	0	0-7	0	OK	1.9	30.000	
explore	mpi	8	4×2	1200	0	0-7	0	OK	1.8	30.000	
explore	mpi	8	8×1	1200	0	0-7	0	OK	1.9	30.000	
explore	openmp	6	1×6	1200	0	0-5	0	OK	2.6	30.000	
explore	openmp	6	2×3	1200	0	0-5	0	OK	2.5	30.000	
explore	openmp	6	3×2	1200	0	0-5	0	OK	2.6	30.000	
explore	openmp	6	6×1	1200	0	0-5	0	OK	2.5	30.000	
explore	openmp	4	1×4	1200	0	0-3	0	OK	3.2	30.000	
explore	openmp	4	2×2	1200	0	0-3	0	OK	3.2	30.000	
explore	openmp	4	4×1	1200	0	0-3	0	OK	3.2	30.000	